

Project 5 – Inventory Management System

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Project Code :

```
1  # Inventory Management System
2
3  inventory = {}
4  transactions = []
5
6
7  def add_product():
8
9      pid = input("Enter Product ID: ")
10
11      if pid in inventory:
12          print("Product already exists")
13          return
14
15      name = input("Enter Product Name: ")
16      category = input("Enter Category: ")
17      price = float(input("Enter Price: "))
18      quantity = int(input("Enter Quantity: "))
19      reorder = int(input("Enter Reorder Level: "))
20
21
22      inventory[pid] = {
23          "name": name,
24          "category": category,
25          "price": price,
26          "quantity": quantity,
27          "reorder": reorder
28      }
29
30      print("Product Added Successfully")
31
32
33
```

```

34 def stock_in():
35
36     pid = input("Enter Product ID: ")
37
38     if pid in inventory:
39
40         qty = int(input("Enter Quantity: "))
41
42         inventory[pid]["quantity"] += qty
43
44         transactions.append(
45             f"IN : {pid} +{qty}"
46         )
47
48         print("Stock Added")
49
50     else:
51         print("Product Not Found")
52
53
54
55 def stock_out():
56
57     pid = input("Enter Product ID: ")
58
59     if pid in inventory:
60
61         qty = int(input("Enter Quantity: "))
62
63
64         if inventory[pid]["quantity"] >= qty:
65
66             inventory[pid]["quantity"] -= qty
67
68             transactions.append(
69                 f"OUT : {pid} -{qty}"
70             )
71
72             print("Stock Removed")
73
74         else:
75             print("Not enough stock")
76

```

```

76
77     else:
78         print("Product Not Found")
79
80
81
82 def view_inventory():
83
84     for pid, p in inventory.items():
85
86         value = p["price"] * p["quantity"]
87
88         print("\nProduct ID:", pid)
89         print("Name:", p["name"])
90         print("Category:", p["category"])
91         print("Price:", p["price"])
92         print("Quantity:", p["quantity"])
93         print("Total Value:", value)
94
95
96
97 def low_stock_alert():
98
99     print("\nLow Stock Products")
100
101     for pid,p in inventory.items():
102
103         if p["quantity"] <= p["reorder"]:
104             print(p["name"], "- Quantity:", p["quantity"])
105
106
107
108 def generate_report():
109
110     total_value = 0
111     categories = set()
112
113
114     for p in inventory.values():
115
116         total_value += p["price"] * p["quantity"]
117         categories.add(p["category"])
118

```

```

120 > print("""===== INVENTORY REPORT ===== ...
122
123     print("Total Products:", len(inventory))
124     print("Total Stock Value:", total_value)
125     print("Categories:", categories)
126
127
128     print("\nTransaction History")
129
130     for t in transactions:
131         print(t)
132
133
134
135 while True:
136
137     print("""
138     ===== INVENTORY MANAGEMENT SYSTEM =====
139
140     1.Add Product
141     2.Stock In
142     3.Stock Out
143     4.View Inventory
144     5.Low Stock Alert
145     6.Report
146     7.Exit
147
148     """)
149
150     choice = int(input("Enter Choice: "))
151
152
153     if choice == 1:
154         add_product()
155
156
157     elif choice == 2:
158         stock_in()
159
160
161     elif choice == 3:
162         stock_out()
163

```

```
150     choice = int(input("Enter Choice: "))
151
152
153     if choice == 1:
154         add_product()
155
156
157     elif choice == 2:
158         stock_in()
159
160
161     elif choice == 3:
162         stock_out()
163
164
165     elif choice == 4:
166         view_inventory()
167
168
169     elif choice == 5:
170         low_stock_alert()
171
172
173     elif choice == 6:
174         generate_report()
175
176
177     elif choice == 7:
178         break
179
180
181     else:
182         print("Invalid Choice")
183
```

Output:

```
PS C:\Users\tanve\Desktop\ITR> c;; cd 'c:\Users\tanve\Desktop\I
ndled\libs\debugpy\launcher' '59752' '--' 'c:\Users\tanve\Desktop

===== INVENTORY MANAGEMENT SYSTEM =====

1.Add Product
2.Stock In
3.Stock Out
4.View Inventory
5.Low Stock Alert
6.Report
7.Exit

Enter Choice:

      1
Enter Product ID: s123
Enter Product Name: Mobile
Enter Category: Electronic
Enter Price: 30000
Enter Quantity: 1
Enter Reorder Level: 1
Product Added Successfully

===== INVENTORY MANAGEMENT SYSTEM =====

1.Add Product
2.Stock In
3.Stock Out
4.View Inventory
5.Low Stock Alert
6.Report
7.Exit

Enter Choice: 4

Product ID: s123
Name: Mobile
Category: Electronic
Price: 30000.0
Quantity: 1
Total Value: 30000.0
```

```
===== INVENTORY MANAGEMENT SYSTEM =====
```

- 1.Add Product
- 2.Stock In
- 3.Stock Out
- 4.View Inventory
- 5.Low Stock Alert
- 6.Report
- 7.Exit

Enter Choice: 4

Product ID: s123
Name: Mobile
Category: Electronic
Price: 30000.0
Quantity: 1
Total Value: 30000.0

```
===== INVENTORY MANAGEMENT SYSTEM =====
```

- 1.Add Product
- 2.Stock In
- 3.Stock Out
- 4.View Inventory
- 5.Low Stock Alert
- 6.Report
- 7.Exit

Enter Choice: 2
Enter Product ID: s123
Enter Quantity: 2
Stock Added

```
===== INVENTORY MANAGEMENT SYSTEM =====
```

- 1.Add Product
- 2.Stock In
- 3.Stock Out
- 4.View Inventory
- 5.Low Stock Alert
- 6.Report

```
6.Report
7.Exit

Enter Choice: 3
Enter Product ID: s123
Enter Quantity: 1
Stock Removed

===== INVENTORY MANAGEMENT SYSTEM =====

1.Add Product
2.Stock In
3.Stock Out
4.View Inventory
5.Low Stock Alert
6.Report
7.Exit

Enter Choice: 5

Low Stock Products

===== INVENTORY MANAGEMENT SYSTEM =====

1.Add Product
2.Stock In
3.Stock Out
4.View Inventory
5.Low Stock Alert
6.Report
7.Exit

Enter Choice: 6
===== INVENTORY REPORT =====

Total Products: 1
Total Stock Value: 60000.0
Categories: {'Electronic'}

Transaction History
IN : s123 +2
```



```
Enter Choice: 6
===== INVENTORY REPORT =====

Total Products: 1
Total Stock Value: 60000.0
Categories: {'Electronic'}

Transaction History
IN : s123 +2
OUT : s123 -1

===== INVENTORY MANAGEMENT SYSTEM =====

1.Add Product
2.Stock In
3.Stock Out
4.View Inventory
5.Low Stock Alert
6.Report
7.Exit

Enter Choice: 7
PS C:\Users\tanve\Desktop\ITR> |
```